

Matgeo Presentation - Problem 3.2.5

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Question

Draw a triangle ABC in which $BC = 6$ cm, $CA = 5$ cm and $AB = 4$ cm.

Solution

→ Let

$$a = \|\mathbf{C} - \mathbf{B}\| = 6\text{cm} \quad (0.1)$$

$$b = \|\mathbf{A} - \mathbf{C}\| = 5\text{cm} \quad (0.2)$$

$$c = \|\mathbf{B} - \mathbf{A}\| = 4\text{cm} \quad (0.3)$$

→ By using cosine law in $\triangle ABC$, we get

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac} \quad (0.4)$$

$$\implies \cos B = \frac{6^2 + 4^2 - 5^2}{2 \times 6 \times 4} \quad (0.5)$$

$$\implies \cos B = \frac{9}{16} \quad (0.6)$$

$$\implies \angle B = \cos^{-1} \left(\frac{9}{16} \right) \approx 55^\circ \quad (0.7)$$

Solution

→ The coordinates of $\triangle ABC$ can then be expressed as

$$\mathbf{A} = c \begin{pmatrix} \cos B \\ \sin B \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \mathbf{C} = \begin{pmatrix} 0 \\ 6 \end{pmatrix}$$

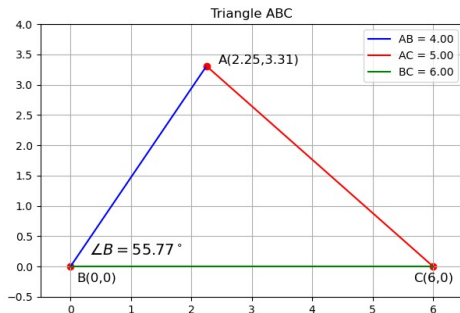


Figure: Plot of $\triangle ABC$

File: points.c

```
#include <stdio.h>

int main() {
    FILE *fp;

    // -----
    // Question 3.2.5
    // -----

    fp = fopen("points.dat", "w");
    fprintf(fp, "%d,%d,%d\n", 2.25, 3.31, 0); // A
    fprintf(fp, "%d,%d,%d\n", 0, 0, 0); // B
    fprintf(fp, "%d,%d,%d\n", 6, 0, 0); // C
    fclose(fp);
    return 0;
}
```

File: call_c.py

```
import subprocess

# Compile the C program
subprocess.run(["gcc", "points.c", "-o", "points"])

# Run the compiled C program
result = subprocess.run(["./points"], capture_output=True, text=True)

# Print the output from the C program
print(result.stdout)
```

File: plot.py

```
import matplotlib.pyplot as plt
import numpy as np

# Given coordinates
B = np.array([0, 0])
C = np.array([6, 0])
A_x = 4 * np.cos(np.deg2rad(55.77)) #  $A_x = 4 * \cos(55.77 \text{ degrees})$ 
A_y = 4 * np.sin(np.deg2rad(55.77)) #  $A_y = 4 * \sin(55.77 \text{ degrees})$ 
A = np.array([A_x, A_y])

# Calculate side lengths
AB = np.linalg.norm(A - B)
BC = np.linalg.norm(C - B)
AC = np.linalg.norm(A - C)

# Calculate angle B using the law of cosines
cos_angle_B = (AB**2 + BC**2 - AC**2) / (2 * AB * BC)
angle_B_rad = np.arccos(cos_angle_B) # In radians
angle_B_deg = np.rad2deg(angle_B_rad) # Convert to degrees

# Create the plot
plt.figure(figsize=(7, 5))

# Plot the triangle
plt.plot([B[0], A[0]], [B[1], A[1]], 'b-', label=f'AB={AB:.2f}')
plt.plot([A[0], C[0]], [A[1], C[1]], 'r-', label=f'AC={AC:.2f}')
plt.plot([C[0], B[0]], [C[1], B[1]], 'g-', label=f'BC={BC:.2f}')
```

File: plot.py

```
# Mark the vertices
plt.scatter([B[0], A[0], C[0]], [B[1], A[1], C[1]], color='red')
plt.text(B[0]+0.1, B[1]-0.1, 'B(0,0)', fontsize=12, ha='left', va='top')
plt.text(A[0]+0.2, A[1], f'A({A_x:.2f},{A_y:.2f})', fontsize=12, ha='left', va='bottom')
plt.text(C[0], C[1]-0.1, 'C(6,0)', fontsize=12, ha='center', va='top')

# Label the angle at B and its value
plt.text(B[0] + 0.3, B[1] + 0.2, r'$\angle B = {:.2f}^\circ$'.format(angle_B_deg), fontsize=14, ha='left')

# Set plot properties
plt.gca().set_aspect('equal', adjustable='box')
plt.xlim(-0.5, 6.5)
plt.ylim(-0.5, 4)

# Title and grid
plt.title('Triangle ABC')
plt.grid(True)

# Show the plot
plt.legend()
plt.show()
```